

On the representation of potential energy surfaces of polyatomic molecules in normal coordinates: II. Parameterisation of the force field

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Abstract

By substituting the standard mass-weighted normal coordinates with either Morse-like or Gauss-like coordinates, it is demonstrated that significant improvements can be made to the vibrational spectra of polyatomic molecules calculated variationally. Quartic force fields in the form of Taylor expansions are generated by density functional theory for water, formaldehyde and methane, and their vibrational spectra calculated by the perturbation normal coordinate code SPECTRO. These are then compared with three sets of spectra arising from the variational code MULTIMODE. Initial spectra are obtained using the identical Taylor expansion force fields. A subsequent set of spectra are then obtained for which the symmetric normal coordinates of the force fields are replaced by Morse-like coordinates and a final set of spectra are obtained for which the asymmetric normal coordinates of the force field are replaced by Gauss-like coordinates. The restriction is imposed that the complete set of derivatives to quartic are preserved under these coordinate transformations.

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1. Introduction

It is well known that quartic force fields expressed as Taylor expansions in displacements from equilibrium fail to produce the correct variational vibrational wavenumbers, especially for stretching motions containing hydrogen atoms [1]. Even though the derivatives of the potential may be *exact*, the variational procedure requires that the hydrogen atoms move in regions of coordinate

space beyond those in which the force field is valid, which means that the force field must be extended. In the usual case for which the quartic force constant is *positive*, the stretching vibrational wavenumbers are calculated *too high* by anything between 20 and 60 cm⁻¹.

In calculations involving internal valence coordinates [1], the usual way round this difficulty is to substitute the stretching coordinates by either Simons–Parr–Finlan coordinates $(R - R_e)/R$ [2] or Morse-like coordinates $1 - e^{-\alpha(R-R_e)}$ [3], where R is the bondlength and R_e is its equilibrium value. These both have the property of dissociating to a

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limit of +1 as R tends to $+\infty$ (for bond displacements the allowed values of R are 0 to $+\infty$). In effect, the radius of convergence has been increased from $2R_e$ to infinity. A result of this substitution is that very good extrapolation of the potential results from the lower-order force constants governing the near-equilibrium regions. For molecules in which two such bond coordinates are linked through symmetry, both symmetric and asymmetric stretches benefit from this substitution since the symmetry force constants are linear combinations of the bond displacement force constants ($F_{ss} = f_{rr} + f_{rr'}$; $F_{aa} = f_{rr} - f_{rr'}$, etc.). In earlier work on water [1], we found that the stretch variational wavenumbers calculated using the Hoy–Mills–Strey [4] Taylor expansion force field (which is merely the transformation of a potential in normal coordinates to one in internal coordinates) were high by about 60 cm^{-1} . Replacing the bond displacement coordinates by Morse-like coordinates, and ensuring that the values of all force constants were preserved, resulted in stretch variational wavenumbers close to the exact values. As is customary, we left the angle-bend coordinate as a displacement from equilibrium.

In variational calculations of the vibrational spectra of small polyatomic molecules, it is now usual to fit polynomials to ab initio energy grids in some convenient internal coordinates. There are two reasons for this. First, the advance of modern electronic structure packages has been so marked that it allows for the calculation of potentials of extreme high quality over wide regions of coordinate space, thus enabling all of the anharmonicity features to be accurately recorded in the potential. Second, this is preferable to using a derivative expansion because there is always the danger, that large displacements from equilibrium may result in unpredictable behaviour of the extrapolated potential. It is also possible that large *holes* may exist in extended regions. Potentials produced ab initio will have calculated energies in these regions and will be almost guaranteed to be well-behaved. However, it is impractical to consider this approach for molecules with more than four atoms.

For larger molecules Taylor expansion force fields in normal coordinates are obtained by differentiating the potential at equilibrium. It is usual

to calculate quartic force fields in this manner. Such force fields are unaware of any gross anharmonicity features in the potential at large displacements. They are also rectilinear in nature, and therefore cannot probe *all* regions of coordinate space successfully.

Our code MULTIMODE is designed to work with normal coordinates, using the variational method. MULTIMODE has been successfully used for calculations of low-lying vibrational levels of benzene using a quartic potential in normal coordinates [5]. However, MULTIMODE is not restricted to quartic normal coordinate force fields. It only requires cuts in the potential in which a maximum of four normal modes are concurrently non-zero, but the form of the cuts can be completely general. In an earlier Letter [6] which forms Part (I) of this Letter, we outlined a method of obtaining accurate normal coordinate potentials for use in MULTIMODE derived from fits to grids of an ab initio data. This followed earlier Letter on the interpolation of the potential between such grids [7]. We now describe a further method for recovering the accuracy lost when working with analytical Taylor expansions of normal coordinate force fields. The common thread between these two approaches is that we recognise that normal coordinates have ranges $-\infty$ to $+\infty$ which can often lead to potentials which behave unpredictably for large displacements. In (I), we introduced the coordinate $\tanh(\gamma Q)$ where Q is a mass-weighted normal coordinate, in order to fit the intrinsic parts of the full $3N - 6$ mode potential. This coordinate has the limits -1 and $+1$ at $Q = -\infty$ and $Q = +\infty$, respectively. We demonstrated that intrinsic potentials from one- to four-modes can be routinely fit to extremely high-accuracy with high-degree polynomials in such coordinates, with no sign of any spurious behaviour in the integration region.

In this Letter, we apply a new transformation procedure (described in Section 2) to mass-weighted normal coordinate force fields. We use density functional theory (DFT) to generate the Taylor expansion force fields, and the variational code MULTIMODE to quantise these potentials. We apply the method to water (three modes), formaldehyde (six modes) and methane (nine modes) in

order to demonstrate the generality of the procedure. We shall compare our energy levels with those obtained from SPECTRO [8], which is a code which applies standard second-order perturbation theory to normal coordinate quartic force fields. If there are no resonances, it is well established in quantum mechanics that predictions from second-order perturbation theory are high quality. Furthermore, it is known that the exact energy levels of a Morse oscillator are obtained by second-order perturbation theory. This comparison of MULTIMODE and SPECTRO will therefore form the basis for our discussion of the benefits of the transformations.

2. Construction and manipulation of the force field

All force fields which are used here were obtained with DFT. Specifically we use B97-1 functional [9] with a TZ2P basis set, with the code CADPAC [10]. The geometry of the molecule was optimised, the second derivative matrix was determined, and the normal coordinates Q_k were obtained, together with the harmonic frequencies ω_k and corresponding vectors $l_{\zeta i,k}$ where i labels the atoms, k the normal coordinates and ζ is one of x, y, z . Higher-order force constants were obtained by finite displacement, specifically

$$F_{ijk} = (F_{ij}(+\delta Q_k) - F_{ij}(-\delta Q_k)) / (2\delta Q_k), \quad (1)$$

$$F_{ijkk} = (F_{ij}(+\delta Q_k) + F_{ij}(-\delta Q_k) - 2F_{ij}(0)) / (\delta Q_k)^2. \quad (2)$$

Here, $F_{ij}(+\delta Q_k)$ is the ij th element of the force constant matrix, in normal coordinates, evaluated at the displaced geometry (equilibrium $+\delta Q_k$), for appropriate values of δ . Thus we have the following representation of the quartic force field:

$$V(\mathbf{Q}) = \frac{1}{2} \sum_i \omega_i^2 Q_i^2 + \frac{1}{6} \sum_{ijk} F_{ijk} Q_i Q_j Q_k + \frac{1}{24} \sum_{ijkl} F_{ijkl} Q_i Q_j Q_k Q_l, \quad (3)$$

where l is constrained to be equal to k . Note that this procedure gives no information for F_{ijkl} when all modes are different.

In all of our sample molecules, there is at least C_{2v} symmetry, and therefore some normal coordinates will be symmetric (A_1) and others will be asymmetric (B_1, A_2, B_2) under operations of this point group. Conversely, this results in potentials which are asymmetric or symmetric about equilibrium, respectively. The corresponding forms of the one-dimensional cuts along these modes will therefore be

$$V(Q_k) = \frac{1}{2} \omega_k^2 Q_k^2 + \frac{1}{6} F_{kkk} Q_k^3 + \frac{1}{24} F_{kkkk} Q_k^4 \quad (4)$$

for all symmetric modes, and

$$V(Q_k) = \frac{1}{2} \omega_k^2 Q_k^2 + \frac{1}{24} F_{kkkk} Q_k^4 \quad (5)$$

for all asymmetric modes. All symmetric modes will therefore exhibit *some* Morse-like behaviour, and following the procedure adopted for stretch valence coordinates, it is possible to replace the coordinate Q_k by the Morse-like coordinate $y_k = 1 - e^{-\alpha_k Q_k}$. In the absence of an experimental dissociation energy, the value of α_k can be obtained directly from the expression

$$\alpha_k = -F_{kkk} / 3\omega_k^2 \quad (6)$$

which arises from assuming that the vibration is associated with a potential

$$V(Q_k) = D_e (1 - e^{-\alpha_k Q_k})^2 \quad (7)$$

and by equating (4) and (7) and expanding $e^{-\alpha_k Q_k}$. As mentioned earlier, Q_k has limits $Q_k = -\infty$ and $Q_k = +\infty$, and since the *sign* of each Q_k is determined from an eigenvector $l_{\zeta i,k}$ of the Wilson FG diagonalisation [11], dissociation can take place for either *positive* or *negative* values of Q_k . The former will give rise to a *negative* value of F_{kkk} , whilst the latter will give rise to a *positive* value of F_{kkk} , respectively. We make an initial scan of the complete normal coordinate force field and pick out the quadratic force constants ω_k^2 , and corresponding cubic force constants F_{kkk} (if present) and form all of the non-zero values of α_k .

For asymmetric modes, there is no cubic force constant F_{kkk} , and therefore the potential climbs monotonically to $V(Q_k) = +\infty$ for $Q_k = \pm\infty$ if F_{kkkk} is *positive*, as is usually the case. Consider however the case of the asymmetric stretch

vibration in water; the limit of this motion is a dissociated atom plus diatom $\text{H} + \text{OH}$, as opposed to the dissociated atoms $\text{H} + \text{O} + \text{H}$ of the symmetric stretch vibration. The point we are making is that even asymmetric vibrations can be associated with *some* dissociation limit, although this is non calculable from the quartic force field. If we assume however that such a vibration can be associated with a potential

$$V(Q_k) = D_e(1 - e^{-\beta_k Q_k^2}), \quad (8)$$

then equating (5) and (8) gives

$$\beta_k = \omega_k^2/2D_e \quad (9)$$

for positive D_e . We therefore replace Q_k by the Gauss-like coordinate $z_k = (1 - e^{-\beta_k Q_k^2})^{1/2}$ for all asymmetric modes. Such a coordinate has a similar parameterisation and flexibility as the Morse-like coordinate. Based on the values of the dissociation energies which *are* calculable for the symmetric modes from (4) and (7), we make an initial guess at those for the asymmetric modes, which allows us to calculate the corresponding values of β_k , which can then be fine-tuned to give an optimum potential.

The remaining task is to form the new force field in terms of y_k and z_k such that all the derivatives of the potential ω_k^2 , F_{ijk} and F_{ijkk} are preserved. For the Morse-like coordinates the required relations are

$$Q_i = (y_i + y_i^2/2 + y_i^3/3 + y_i^4/4)/\alpha_i, \quad (10)$$

$$Q_i^2 = (y_i^2 + y_i^3 + 11y_i^4/12)/\alpha_i^2, \quad (11)$$

$$Q_i^3 = (y_i^3 + 3y_i^4/2)/\alpha_i^3, \quad (12)$$

$$Q_i^4 = y_i^4/\alpha_i^4. \quad (13)$$

We rewrite the normal coordinate potential (3) in a general form

$$V(\mathbf{Q}) = \sum_i G_i Q_i + \sum_{i \leq j} G_{ij} Q_i Q_j + \sum_{i \leq j \leq k} G_{ijk} Q_i Q_j Q_k + \sum_{i \leq j \leq k} G_{ijkk} Q_i Q_j Q_k^2, \quad (14)$$

where $G_i = G_{i < j} = 0$ and $G_{ii} = \frac{1}{2}\omega_i^2$, etc., in (3). The force constants for the corresponding one-dimensional cuts in y_i are $(V(\mathbf{y}) = \sum_i f_i y_i + \dots)$

$$f_i = G_i/\alpha_i, \quad (15)$$

$$f_{ii} = G_{ii}/\alpha_i^2 + G_i/2\alpha_i, \quad (16)$$

$$f_{iii} = G_{iii}/\alpha_i^3 + G_{ii}/\alpha_i^2 + G_i/3\alpha_i, \quad (17)$$

$$f_{iiii} = G_{iiii}/\alpha_i^4 + 3G_{iii}/2\alpha_i^3 + 11G_{ii}/12\alpha_i^2 + G_i/4\alpha_i. \quad (18)$$

Therefore the cubic force constant G_{ijk} will contribute to the Morse-like constants f_{ijk} , f_{iijk} , f_{ijjk} and f_{ijkk} . Note that in the new coordinates we work with a quartic force field with again only the semi-diagonal quartic terms.

For the Gauss-like coordinates z_i the relations are

$$Q_i^2 = (z_i^2 + z_i^4/2)/\beta_i, \quad (19)$$

$$Q_i^4 = z_i^4/\beta_i^2, \quad (20)$$

$$f_{ii} = G_{ii}/\beta_i, \quad (21)$$

$$f_{iiii} = G_{iiii}/\beta_i^2 + G_{ii}/2\beta_i. \quad (22)$$

Therefore the cubic force constant G_{iik} will contribute to the Gauss-like constants f_{iik} and f_{iikk} .

In the following Sections, we use the Taylor expansion force fields in (3) in order to obtain the vibrational spectra of water, formaldehyde and methane. These are first obtained directly from the force fields by SPECTRO, which uses standard second-order perturbation theory to calculate the spectra. We again use these same force fields to calculate the spectra, this time by MULTIMODE, which is a near-exact variational procedure using the exact Watson normal coordinate hamiltonian [12]. MULTIMODE first performs a vibrational SCF calculation in order to optimise the one-dimensional harmonic-oscillator basis, and this is followed by a CI using the virtual functions for the zero-point level. Its approximation is that it integrates over grids of normal coordinates to a maximum of *four* coupled modes only [13]. This however allows the calculation of energy levels for molecules containing considerably more than four modes [5], but in this case there is an increasing overhead in the size of the CI matrix to be diagonalised. Schemes to tailor the CI matrix to the

molecules have therefore been introduced [14]. Input into MULTIMODE is the potential, which is ultimately converted to a potential in the cuts of the normal coordinates. We modify the force fields according to Eqs. (10)–(13), by replacing the normal coordinates Q_s with the Morse-like coordinates y_s for all symmetric modes, and we recalculate the MULTIMODE spectra. Finally, we modify the force fields further according to Eqs. (19) and (20), by replacing the normal coordinates Q_a with the Gauss-like coordinates z_a for all asymmetric modes, and we again recalculate the spectra.

3. Results for water

The calculated spectra of water are given in Table 1. We comment on the various columns in turn. The initial results for SPECTRO did not include the effect of Darling–Dennison interactions between $2\nu_1$ and $2\nu_3$. We have therefore calculated this effect independently, and found it to be 31.09 cm^{-1} . This has been used to compile the spectrum in Table 1. The first set of MULTIMODE results (MM-T) employed the identical Taylor expansion force field in Eq. (3) as that used by SPECTRO. As with internal coordinates [1], the variational spectrum gives both symmetric (ν_1) and asymmetric (ν_3) stretch wavenumbers too *high*, by about 40 cm^{-1} , which is however slightly less than that for the valence calculations (60 cm^{-1}). The corresponding errors in the stretch overtones are 168 and 143 cm^{-1} , respectively. Since there are

only three modes in water, the MULTIMODE results correspond to *complete* integration, and the basis set used ensures virtually complete convergence. It follows that for a potential of this type these correspond to definitive variational results, and hence are of little value, since SPECTRO is known to give superior results for accurate force fields (as discussed in Section 1 [4]).

The next column contains the MULTIMODE results (MM-M) for a potential in which only the normal coordinates for the symmetric modes have been replaced by Morse-like coordinates according to Eqs. (10)–(13). The immediate result is that the discrepancy in ν_1 has been almost totally removed by this transformation. In fact, without exception, all levels show an improvement from those of the Taylor expansion potential, most notably that for the asymmetric stretch ν_3 , which is however, still 20 cm^{-1} too high. We anticipated this earlier, when we noted that Morse-like coordinates could not be applied to asymmetric modes due to the lack of the cubic force constant F_{kkk} . It is difficult to assess the effect of second-order effects, especially those involving $2\nu_1$ and $2\nu_3$, but it can be seen that the errors in these levels have been reduced to 30 and 60 cm^{-1} , respectively.

The final column contains MULTIMODE results (MM-MG) where, in addition to the Morse-like coordinates, we replace the normal coordinate for the asymmetric stretch mode by a Gauss-like coordinate according to Eqs. (19) and (20). We tried several values of β and found that the results were almost invariant for a wide range (corresponding to a large range of values of an associated

Table 1
Low lying vibrational levels of water, calculated using SPECTRO and MULTIMODE

Symmetry	Level	SPECTRO	MM-T	MM-M	MM-MG
A_1	zpe	4642.41	4652.46	4645.83	4642.13
	ν_2	1595.45	1586.01	1588.85	1592.01
	$2\nu_2$	3156.05	3134.82	3140.97	3157.14
	ν_1	3672.25	3712.33	3675.59	3666.09
	$2\nu_1$	7231.16	7399.04	7263.01	7229.36
	$2\nu_3$	7465.03	7608.30	7532.29	7467.87
B_2	ν_3	3763.66	3810.01	3781.62	3764.62

The calculations using MULTIMODE involve three distinct potentials: (i) Taylor expansion (MM-T) of the normal coordinates; (ii) Morse-like expansion (MM-M) of symmetric normal coordinates; (iii) optimised Morse-like plus Gauss-like expansions (MM-MG) of symmetric and asymmetric normal coordinates.

dissociation energy D_e). However it can be seen that the SPECTRO results are well represented by our final choice of combined Morse–Gauss values, as is clearly demonstrated by looking at the mean absolute difference of the MULTIMODE results from the SPECTRO results; for MM-T it is 63 cm^{-1} , for MM-MG it is 2 cm^{-1} .

4. Results for formaldehyde

The calculated spectra of formaldehyde are given in Table 2. Here, the original SPECTRO results have been augmented by Darling–Dennison resonances between $2\nu_4$ and $2\nu_6$, and between $2\nu_1$ and $2\nu_5$. These shifts amount to 1.44 cm^{-1} for $2\nu_4$ and $2\nu_6$ and 44.53 cm^{-1} for $2\nu_1$ and $2\nu_5$. The results also include Fermi I resonances between $2\nu_3$ and ν_1 ; Fermi II resonances between $\nu_1 + \nu_6$ and ν_5 , and between $\nu_3 + \nu_6$ and ν_5 , all calculated internally by SPECTRO. A comparison between the SPECTRO results and the first set of MULTIMODE results (MM-T) for the Taylor expansion potential indicates slightly less long-range gross anharmonicity than in water. However, the symmetric (ν_1) and antisymmetric (ν_5) stretch wavenumbers are both *high*, by 16 cm^{-1} ; the corresponding errors in the stretch overtones are 98 and 122 cm^{-1} , respec-

tively. Levels associated with angular motion are very similar to each other.

The MULTIMODE results (MM-M) for the Morse-like potential and those (MM-MG) for the combined Morse–Gauss potential are themselves very similar, and the MM-M values almost fully recover the bulk of the errors found in the MM-T values. The MM-MG results are however, slightly more consistent with the SPECTRO numbers. The reduction in anharmonicity is reflected in the mean absolute differences between the SPECTRO results and the MULTIMODE results; for MM-T it is 23 cm^{-1} , whilst for MM-MG it is 5 cm^{-1} . These are more similar to each other than the corresponding values for water.

One important feature exists in water that is absent in formaldehyde (and methane), and that is its ability to assume linear structures relatively easily ($10\,000\text{ cm}^{-1}$). The shape of the bending potential in water is therefore very Morse-like towards linearity, which must correspondingly have an effect on the remaining rectilinear normal coordinates. We believe that this is one major contributor to the overall anharmonicity effects that appear in our water spectrum, but which is mainly absent in the formaldehyde spectrum. The dissociative features of the hydrogen atoms are however present in both spectra as anticipated.

Table 2

Low lying vibrational levels of formaldehyde, calculated using SPECTRO and MULTIMODE

Symmetry	Level	SPECTRO	MM-T	MM-M	MM-MG
A_1	<i>zpe</i>	5741.46	5751.50	5748.65	5749.04
	ν_3	1499.14	1494.97	1495.91	1499.07
	ν_2	1797.08	1797.89	1796.84	1797.32
	$2\nu_4$	2352.04	2344.69	2342.92	2342.81
	$2\nu_6$	2486.23	2479.74	2479.75	2479.75
	ν_1	2743.47	2759.72	2744.66	2743.15
	$2\nu_3$	2998.69	2987.87	2989.83	3001.35
	$2\nu_2$	3576.51	3580.80	3575.19	3576.28
	$2\nu_1$	5389.02	5487.69	5420.65	5412.62
	$2\nu_5$	5529.19	5651.61	5520.89	5524.50
B_2	ν_6	1244.59	1240.59	1241.81	1242.48
	ν_5	2823.86	2838.73	2828.94	2829.66
B_1	ν_4	1180.68	1175.86	1177.25	1178.09

The calculations using MULTIMODE involve three distinct potentials: (i) Taylor expansion (MM-T) of the normal coordinates; (ii) Morse-like expansion (MM-M) of symmetric normal coordinates; (iii) optimised Morse-like plus Gauss-like expansions (MM-MG) of symmetric and asymmetric normal coordinates.

5. Results for methane

The calculated spectra of methane are given in Table 3. Before discussing the various spectra, we recall that methane has T_d symmetry, and therefore there are doubly degenerate (E) and triply degenerate (T_2) vibrational levels. However, MULTIMODE only works in non-degenerate point groups. In C_{2v} symmetry the E species become A_1 and A_2 , and the T_2 species become A_1 , B_1 and B_2 . Because of this, the different components of E and T_2 are not equivalently represented by the C_{2v} basis in MULTIMODE, and must be converged to degeneracy. Furthermore, as noted earlier [15], the lack of five-mode coupling also results in only two components of T_2 achieving complete degeneracy, and this effect is reproduced here. The error is only a few cm^{-1} , which is not sufficient to interfere with the objective of the present investigation. We note in passing, however, that SPECTRO exhibits a similar behaviour for both E and T_2 levels, but with differences between only 1 and 2 cm^{-1} for the fundamentals. This is not the same

effect, but is due to the fact that the normal co-ordinate eigenvectors are not strictly irreducible representations of T_d . These effects are therefore transferred to MULTIMODE, as can be seen in all cases except ν_3 , where the combination of both effects increases the error to 8 cm^{-1} . If it was possible to include full symmetry, in the MM-T column the $\nu_4(T)$ levels at 1297.97, 1300.33, and 1300.39 would all be strictly degenerate. Although SPECTRO can operate in higher symmetry, we considered it appropriate to work in C_{2v} symmetry throughout. The consequence of this is that the SPECTRO overtones are not correct, because the nearly degenerate levels should mix.

A further point to note for methane is that, during the determination of the Morse-like parameters α_k , all except two were determined to be close to zero. This implies a cubic force constant F_{kkk} close to zero, or in other words a more harmonic potential. In these cases, the Morse-like coordinate y_k is proportional to Q_k (as can be seen by expanding y_k) and we therefore maintain the coordinate Q_k in our calculations. Similarly, since

Table 3
Low lying vibrational levels of methane, calculated using SPECTRO and MULTIMODE, and compared with experiment

Symmetry	Level	SPECTRO	MM-T	MM-M	Expt. [15]
A_1	zpe	9625.83	9635.61	9633.45	
	$\nu_4(T)$	1304.24	1297.97	1298.13	1310.76
	$\nu_2(E)$	1523.49	1518.77	1518.84	1533.33
	$2\nu_4$	2599.73	2570.62	2568.16	2587.04
	$2\nu_4(T)$	2600.17	2595.21	2594.81	2614.26
	$2\nu_4(E)$	2600.21	2601.91	2601.93	2624.62
	ν_1	2902.58	2915.33	2907.31	2916.48
	$\nu_3(T)$	2997.64	3013.25	3008.70	3019.49
	$2\nu_2$	3047.73	3034.38	3034.68	3063.65
	$2\nu_2(E)$	3053.93	3041.10	3041.19	3065.14
	$2\nu_1$	5781.65	5811.47	5781.59	5790.00
	$2\nu_3$	5936.79	5995.14	5985.21	5968.10
	$2\nu_3(T)$	5936.99	6038.48	6013.33	6004.60
	$2\nu_3(E)$	5966.74	6042.67	6032.57	6043.80
B_2	$\nu_4(T)$	1306.33	1300.33	1300.43	1310.76
	$\nu_3(T)$	2998.51	3021.15	3016.64	3019.49
B_1	$\nu_4(T)$	1306.31	1300.39	1300.49	1310.76
	$\nu_3(T)$	2998.61	3021.23	3016.59	3019.49
A_2	$\nu_2(E)$	1525.83	1521.31	1521.33	1533.33

The calculations using MULTIMODE involve two distinct potentials: (i) Taylor expansion (MM-T) of the normal coordinates; (ii) Morse-like expansion (MM-M) of symmetric normal coordinates. Experimental values are included to demonstrate the good prediction of MM-M for the splitting of the overtones. The overtones from SPECTRO are not reliable (see text).

we determine the Gauss-like parameter β_k in a similar way, we maintain the coordinate Q_k where the assumed dissociation energy is high. A few test runs suggested that *all* asymmetric coordinates may be left as the normal coordinates Q_k with little change to the calculated spectra. The column MM-MG is therefore absent from Table 3.

In Table 3, we give values for the zero point energy, together with the wavenumbers of the fundamentals and those first overtones with A_1 symmetry. The reason for this is that there exist many resonances in methane which are not explicitly calculated by SPECTRO but which are automatically calculated by MULTIMODE. This makes a direct comparison between SPECTRO and MULTIMODE very difficult in many cases where multiple quanta are involved. For this reason, we supplement the table with the corresponding experimental values. We emphasise that this is *not* to compare the accuracy of the force fields, but to give a rough estimate of the size of the resonances mentioned above.

We commence with the fundamentals which are resonance-free, to all intents and purposes. In all cases, the Morse-like levels under MM-M are closer to the SPECTRO levels than are those of the Taylor expansion under MM-T, with the two CH stretching vibrations (ν_1 and ν_3) undergoing the major improvement as expected, although not as pronounced as for water or formaldehyde for this somewhat more harmonic potential. Turning now to the first overtones, there are resonances involving all of them, so we will go through them in turn. For $2\nu_4$ the three components, degenerate in zero-order, will be split by a strong Darling–Dennison-type resonance, and taking the experimental differences as a guide, these splittings amount to 27 and 10 cm^{-1} , compared with 26 and 7 cm^{-1} for MM-M, respectively. For $2\nu_2$ the two degenerate zero-order components are again split by a similar resonance, and this is shown to be 2 cm^{-1} from experiment and 7 cm^{-1} from MM-M. The overtones $2\nu_1$ and $2\nu_3$ are very complicated, as there is also a Darling–Dennison resonance between these levels, and between the three degenerate components of $2\nu_3$. From experiment, the resulting differences/splittings are 178 cm^{-1} for $2\nu_1$ and $2\nu_3$, and 36 and 39 cm^{-1} for the three com-

ponents of $2\nu_3$. The corresponding MM-M values are 204 cm^{-1} for $2\nu_1$ and $2\nu_3$, and 28 and 19 cm^{-1} for the three components of $2\nu_3$, respectively.

The differences between MM-T and MM-M for the overtones are again shown to be mainly for levels involving ν_1 and ν_3 , which is consistent with the observations made for the fundamentals. We cannot take the comparison with experiment any more seriously than that of a guide only, as the Fermi and Darling–Dennison resonances depend on the accuracies of cubic and quartic force constants, and our DFT potentials are not expected to be highly accurate.

6. Conclusion

In this Letter we have introduced the effect of long-range anharmonicity into conventional Taylor expansion force fields, expressed in mass-weighted normal coordinates. For modes which are totally symmetric (in any point group) this is done by replacing the normal coordinate Q_k by an equivalent Morse-like coordinate $y_k = 1 - e^{-\alpha_k Q_k}$; for all other modes we replace Q_k by an equivalent Gauss-like coordinate $z_k = (1 - e^{-\beta_k Q_k^2})^{1/2}$. We have shown that the overestimate of the vibrational wavenumbers calculated variationally with the Taylor expansion force fields, in particular those associated with stretching motions involving hydrogen, are corrected by these substitutions in which the Taylor expansion force constants are fully conserved to quartic. We believe that this will prove to be a useful tool for large polyatomic molecules for which more extensive global potentials become prohibitive. We reach this conclusion since our variational results closely resemble those obtained from SPECTRO using the original Taylor expansion force field, for which it is known that, in the absence of resonances, the results are close to experiment if the force field is itself accurate. Our variational results are obtained from MULTIMODE, which is a near-exact procedure in which *all* resonances are automatically included. It follows therefore that if the modified potentials, derived from accurate force fields, reproduce the SPECTRO results in the absence of resonances, the fully resonating spectrum will also be accurate.

Such a claim can never be made by perturbation theory procedures.

If a Taylor expansion is used in a variational calculation at extreme long range, there will always be some spurious behaviour at some stage, and this will be fatal if the high-order constants are negative. These *holes* are usually the result of high-order cross-terms, and the transformations plus variational calculations presented here overcome this in two ways. First, we have demonstrated that this uncertain behaviour occurs at longer range for the Morse–Gauss modified potential than for the Taylor expansion. This is because there is a greater range of accurate extrapolation associated with the Morse–Gauss potentials than for pure displacements; although at infinity the sums of all coefficients will define the *dissociation* limit, which will not be realistic. Second, MULTIMODE first obtains a contracted one-dimensional basis from one-dimensional cuts in the potential on Gauss quadrature points. The contracted basis is then used on HEG points [16], which lie *within* the Gauss points. The basis functions therefore have zero amplitude at extreme long range, so any *hole* will be ignored.

The zero point energy, as computed by SPECTRO, only includes contributions from the quartic potential in normal coordinates. On the other hand the zero point energy from MULTIMODE is the lowest eigenvalue, and therefore must use the full force field. Therefore the zero point energies, calculated by these two methods, can never be the same.

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